# A Hierarchical Bayesian Audit of Step-Counting Algorithm Error Across Sex and Body Geometry

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## 1 Motivation

Criado Perez [2019] frames the gender data gap as inattention rather than malice: products designed around male bodies quietly work less well for everyone else. Wearable step-counting is a clean test bed for the idea, since the algorithm is a single operation (count peaks in an accelerometer signal) whose accuracy plausibly depends on the body geometry of the person wearing it. I wear an Oura ring daily, which is what made the question concrete enough to want to answer with a model rather than a hunch.

Prior work points in the right direction without resolving it. Crouter et al. [2005] showed spring-levered pedometer accuracy degrades with body mass index, mediated by pedometer tilt. Sequeira et al. [1995] reported sex-disaggregated step counts in a large survey without modeling the measurement process. Abel et al. [2011] tested sex and leg length as predictors of step-count error across five motion sensors in 59 adults and found no effect; this is the near-neighbor frequentist null that the present analysis refines with a Bayesian bound. Rowe et al. [2011] established that walking cadence varies by more than 20 spm across the adult height range, which sets a clear mechanistic prior: if body size matters at all, the effect should be visible. Straczkiewicz et al. [2023a] called their own classifier "one-size-fits-most," an honest acknowledgement that subgroup performance is open. None of these studies puts a credible interval on a demographic effect on algorithm error against motion-capture ground truth, with body geometry decomposed into measured covariates.

This project does that. Four nested hierarchical Bayesian models are fit on per-trial cadence error from a naive fixed-threshold step detector, against synchronized force-plate and motion-capture ground truth, on the publicly available Kuopio Gait Dataset [Lavikainen et al., 2024]. Bayesian partial pooling is the right tool for a small unbalanced cohort (17 female, 30 male): it handles the imbalance without manufacturing data, and the credible interval is the deliverable.

**Research question.** How does per-trial cadence error from a naive pelvis-IMU step detector vary with sex and measured body geometry (leg length, inter-ASIS distance, mass), and how tightly can the credible intervals constrain that variation in the available cohort?

## 2 Data

The Kuopio Gait Dataset [Lavikainen et al., 2024] provides synchronized inertial measurement unit (IMU) and optical motion-capture recordings of barefoot overground walking at three instructed speeds (slow, comfortable, fast). Ground truth is derived from Vicon motion capture and floor-embedded force plates. Anthropometrics are caliper-measured (leg length, inter-ASIS distance, mass), not self-reported.

The analysis cohort is the 47 subjects with usable IMU recordings (17 female, 30 male; 36% female), after excluding subjects 11, 14, 37, and 49 per the dataset documentation's missing-IMU annotation. Demographics are in Table 1. The cohort is anthropometrically more overlapping than a representative adult population: between-sex differences in leg length (3.4%) and mass (15%) are both smaller than 1 SD of the

pooled distribution. Inter-ASIS distance runs the opposite direction (F mean 239.5 mm, M mean 235.9 mm), so it carries information beyond what sex alone predicts.

| Metric | F ($n$=17) mean (sd) | M ($n$=30) mean (sd) | All ($n$=47) mean (sd) | All min–max |
|---|---|---|---|---|
| Age (years) | 28.0 (7.2) | 29.3 (5.1) | 28.8 (5.9) | 20–45 |
| Height (cm) | 169.0 (5.0) | 177.5 (6.1) | 174.4 (7.0) | 161.0–188.8 |
| Mass (kg) | 69.6 (6.8) | 80.3 (16.0) | 76.4 (14.3) | 54.1–136.1 |
| Leg length (mm) | 803.7 (42.9) | 831.2 (41.0) | 821.3 (43.4) | 739–893.5 |
| Inter-ASIS (mm) | 239.5 (11.9) | 235.9 (19.5) | 237.2 (17.1) | 193–285 |

Table 1: Cohort demographics for the 47 modeling-eligible subjects. Anthropometrics are caliper-measured; mass is from a calibrated laboratory scale. The All-row means are weighted averages over the 17F/30M split. Source: `report_resources/cohort_demographics_summary.csv`.

**Trial-level exclusions.** Of 2,818 attempted walking trials, 37 (1.31%) were excluded prior to modeling, almost entirely for fewer than 3 force-plate strikes ($n_{\text{strikes}} < 3$) at fast speeds where stride length exceeded the plate array's coverage. Exclusions skew male (1.78% of male trials vs. 0.49% of female trials, F:M ratio 0.28) and concentrate in one male subject (subject 10 alone accounts for 13 of 37). The included data slightly over-represents female trials relative to attempts, so there is no selection bias against the minority subgroup. The modeling cohort is $N = 2{,}781$ trials across all 47 subjects.

# 3 Pipeline and outcome

The pelvis IMU is sampled at 100.1 Hz and recorded continuously across each trial; force plates are sampled at 1 kHz and register strikes only while a foot is in contact with the plate array. Heel-strike candidates from the IMU are detected from the magnitude of free acceleration via a fixed-threshold peak finder (`scipy.signal.find_peaks`, prominence 0.3 g, minimum spacing 30 samples corresponding to a 200 spm upper bound). The detector is deliberately generic: this is an audit of an off-the-shelf threshold-based algorithm, not a validation of a tuned one. Force-plate ground-truth strikes are extracted via a 4th-order Butterworth low-pass filter at 20 Hz applied bidirectionally for zero-phase, then threshold crossings on vertical force.

**Outcome variable.** The per-trial outcome is cadence error,

$$y_{ij} \;=\; \text{cadence}_{\text{IMU},\,ij} - \text{cadence}_{\text{plate},\,ij} \qquad \text{(steps per minute)},$$

where $i$ indexes subject and $j$ indexes trial within subject. Cadence (rather than total step count) is the modeled outcome because the IMU and force plates observe different time windows by design: the IMU records continuously across the full trial (typically ∼10 s) while the force plates only register the ∼1 s during which the subject's feet land on the plate array. A raw count comparison therefore conflates window length with detector accuracy, while cadence is a rate that is well defined on any window containing at least two strikes. Figure 1 demonstrates the window-mismatch problem and the rate-invariance argument on a single representative trial.

# 4 Models

Four nested hierarchical Bayesian models are fit, each adding one layer of structure over the previous, all in PyMC [Abril-Pla et al., 2023] with NUTS. All four share a likelihood family and an observation-scale specification arrived at via posterior predictive checks on Models 1 and 2 (§4.2). Models are compared by

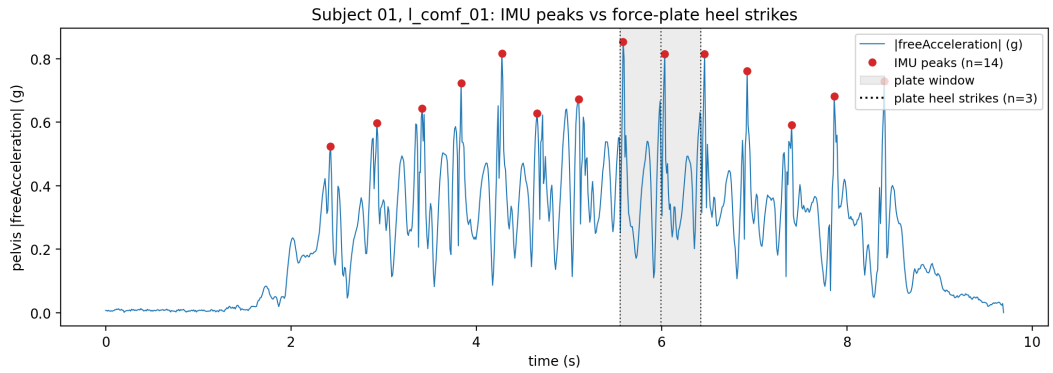


Figure 1: Single-trial detector calibration on subject 01, trial `l_comf_01`. The pelvis IMU records continuously across the full ∼9.5 s trial and the naive fixed-threshold peak detector finds 14 strikes; the five force plates only register the ∼1 s window during which the subject's feet land on the plate array, yielding 3 ground-truth heel strikes. Raw step counts disagree by nearly 5×, almost entirely a measurement-geometry artifact. Cadence is invariant to window length: 130.6 spm from the IMU vs. 137.3 spm from the plate, a −4.9% difference inside the ±10% calibration tolerance, on the first run with no detector tuning. Cadence is therefore the modeled outcome.

leave-one-out cross-validation in §6. All scale parameters use weakly informative `HalfNormal` priors rather than the course's BUGS-convention precision-on-Gamma idiom; with multiple coupled scale parameters in the hierarchical layers, vague Gamma-on-precision priors are known to produce sampler pathologies. Continuous covariates are 1-SD standardized per Gelman [2008].

## 4.1 Model specifications

Let $i$ index trials and $s = s(i)$ index the subject of trial $i$; let $g(s) \in \{F, M\}$ index the sex of subject $s$.

**Model 1 (fully pooled):** a single population mean and observation scale, no subject or sex structure.

$$y_i \sim \mathcal{N}(\alpha,\ \sigma^2_{\text{obs}}), \qquad \alpha \sim \mathcal{N}(0, 10^2), \qquad \sigma_{\text{obs}} \sim \text{HalfNormal}(10).$$

**Model 2 (subject-level partial pooling, cohort observation scale):** each subject gets their own intercept, partially pooled toward the population mean.

$$y_i \sim \mathcal{N}(\alpha_s,\ \sigma^2_{\text{obs}}), \qquad \alpha_s \sim \mathcal{N}(\mu_{\text{pop}},\ \sigma^2_{\text{subj}}),$$

with $\sigma_{\text{subj}} \sim \text{HalfNormal}(5)$ and $\sigma_{\text{obs}} \sim \text{HalfNormal}(10)$. Speed dummies (slow, fast; comfortable as reference) enter at the trial level with $\beta_{\text{slow}}, \beta_{\text{fast}} \sim \mathcal{N}(0, 5^2)$.

**Model 3 (sex as group-level predictor):** subjects pool toward a sex-specific mean; sex-specific means pool toward an overall mean.

$$\alpha_s \sim \mathcal{N}(\alpha_{\text{sex},\,g(s)},\ \sigma^2_{\text{subj}}), \qquad \alpha_{\text{sex},\,g} \sim \mathcal{N}(\mu_{\text{pop}},\ \sigma^2_{\text{sex}}),$$

with $\sigma_{\text{sex}} \sim \text{HalfNormal}(5)$. The sex contrast is tracked as a Deterministic node,

$$\Delta_{F-M} \ = \ \alpha_{\text{sex},\,F} - \alpha_{\text{sex},\,M}.$$

**Model 4 (sex plus body geometry):** Model 3 augmented with 1-SD-standardized subject-level anthropometric covariates $X_s = (\text{leg}_z,\ \text{IAD}_z,\ \text{mass}_z)$:

$$\alpha_s \sim \mathcal{N}\big(\alpha_{\text{sex},\,g(s)} + X_s^\top \boldsymbol{\beta},\ \sigma^2_{\text{subj}}\big), \qquad \boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0},\ 5^2 I).$$

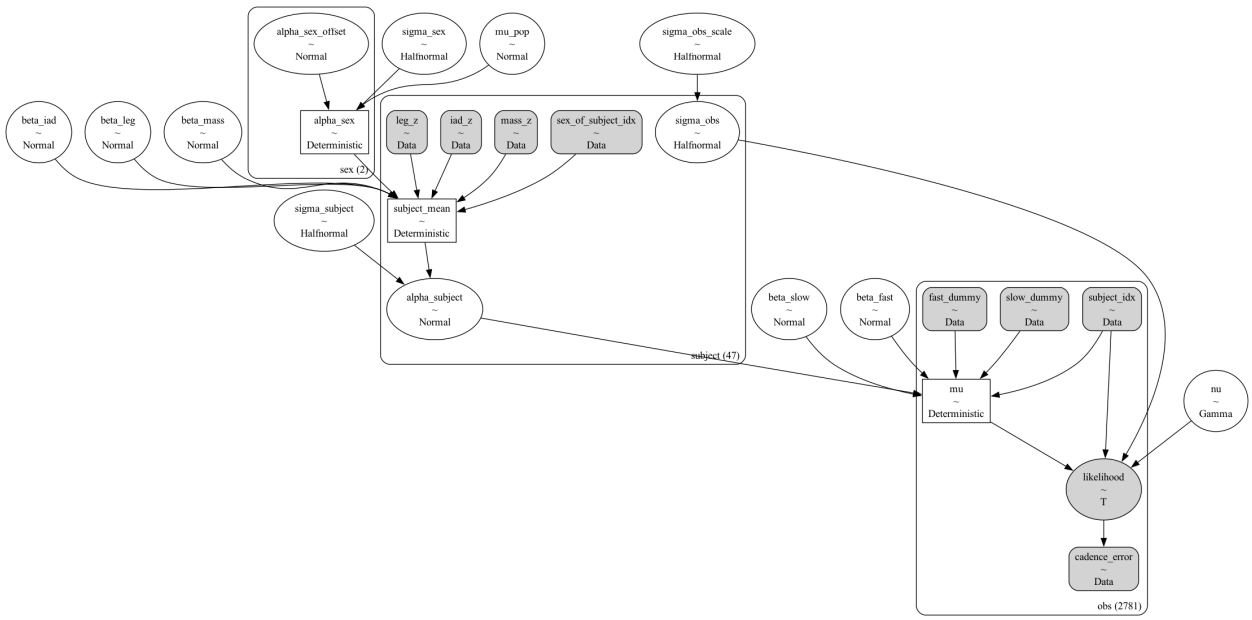


Figure 2: Model 4 directed acyclic graph (`pm.model_to_graphviz`). Three nested plates: sex ($n$=2), subject ($n$=47), observation ($n$=2,781 trials). The Student-t likelihood and per-subject observation scale $\sigma_{\text{obs}}[s]$ are inherited from the specification described in §4.2. $\alpha_{\text{sex}}$ is non-centered via $\alpha_{\text{sex,offset}}$; $\alpha_{\text{subj}}$ is centered. Anthropometric predictors enter at the subject level via $\beta_{\text{leg}}$, $\beta_{\text{iad}}$, $\beta_{\text{mass}}$ on z-scored covariates; speed dummies enter at the trial level. Shaded nodes are data; ellipses are stochastic; rectangles are deterministic.

The geometry-adjusted sex contrast $\Delta^{\text{resid}}_{F-M} = \alpha_{\text{sex},\,F} - \alpha_{\text{sex},\,M}$ is tracked separately from Model 3's raw $\Delta_{F-M}$. The comparison decomposes the sex effect into a geometry-mediated component (the shrinkage between the two posteriors) and a residual component (the Model 4 posterior itself). Figure 2 shows the Model 4 DAG.

## 4.2 Likelihood and observation scale

The Normal-likelihood, cohort-$\sigma_{\text{obs}}$ specification above is the starting point. Two posterior predictive failures on Models 1 and 2 (Figure 5 in the Appendix) prompted two specification changes, both of which propagate to Models 3 and 4 unchanged.

**Likelihood: Normal → Student-t.** Model 1's posterior predictive ECDF exited the 200-draw band on both tails, with a deep left tail near −50 spm and a milder right-tail overshoot. Refitting with a Student-t likelihood, $y_i \sim t_\nu(\alpha, \sigma_{\text{obs}})$ with $\nu \sim \text{Gamma}(2, 0.1)$, tightened the central fit and improved trial-level LOO ($\Delta$ELPD = 871, dse = 53). A skew-t was attempted to address the residual left-shoulder asymmetry; the Fernández–Steel parameterization in the current PyMC release could not be regularized via Gamma priors without leaking prior-predictive mass beyond biomechanically plausible ranges, so Student-t was adopted as the final likelihood. The residual left-shoulder gap is named in Limitation 6 as an unmodeled mixture process.

**Observation scale: cohort → per-subject.** Adding subject partial pooling alone (Model 2, cohort $\sigma_{\text{obs}}$) did not reshape the predictive tails. Posterior $\nu$ actually decreased relative to Model 1-t, indicating the fat tails are trial-level rather than subject-level. Re-specifying the observation scale as per-subject under a

hierarchical hyperprior,

$$\sigma_{\text{obs}}[s] \sim \text{HalfNormal}(\sigma_{\text{obs,scale}}), \qquad \sigma_{\text{obs,scale}} \sim \text{HalfNormal}(10),$$

tightened both tails substantially (Model 2b). A control fit with per-subject $\sigma_{\text{obs}}[s]$ but a Normal likelihood (Model 2a) overfit ($p_{\text{loo}} = 228$ against $\sim$99 parameters) and was rejected. Student-t and per-subject observation scale are both needed; neither alone is sufficient. Models 3 and 4 inherit this specification.

**Sampling.** All models were fit with 4 chains, 2,000 tune plus 2,000 draws per chain (8,000 post-tune total), `target_accept = 0.995`, and `idata_kwargs` configured to log per-observation likelihood for LOO. The sex-level intercept $\alpha_{\text{sex}}$ in Models 3 and 4 is non-centered via an auxiliary offset; the subject-level intercept $\alpha_{\text{subj}}$ is centered, since a non-centered subject layer made sampling worse. The 47-subject layer has enough data per group to favor centered geometry.

# 5 Diagnostics

All five model fits converged cleanly: $\hat{R} = 1.00$ and bulk effective sample size above 3,170 across all population-level parameters in every model (Table 2). The bulk-ESS bottleneck in Models 3 and 4 is $\sigma_{\text{sex}}$, whose 95% HDI lower bound sits at zero against its HalfNormal(5) prior. This is expected behavior for a 2-group hierarchical scale: with only two sex groups, partial pooling cannot estimate the between-sex SD independently of its prior. The parameter is reported with that caveat rather than treated as a failure.

Divergences number at most 3 of 8,000 post-tune draws across all four reporting models, after the sampler reparameterization described above.

| Model | max $\hat{R}$ | min $\text{ESS}_{\text{bulk}}$ | Divergences (of 8,000) |
|---|---|---|---|
| Model 1 (Normal, fully pooled) | 1.00 | 3,469 | 0 |
| Model 1-t (Student-t, fully pooled) | 1.00 | 4,858 | 0 |
| Model 2b (Student-t + per-subject $\sigma$) | 1.00 | 4,305 | 0 |
| Model 3 (+ sex) | 1.00 | 3,172 | 2 (0.025%) |
| Model 4 (+ anthropometrics) | 1.00 | 3,696 | 3 (0.038%) |

Table 2: Convergence diagnostics across all population-level parameters. $\text{ESS}_{\text{bulk}}$ bottleneck in Models 3 and 4 is the prior-dominated $\sigma_{\text{sex}}$.

# 6 Model comparison

Models are compared by PSIS-LOO-CV at two grains. Trial-grain LOO treats each of the 2,781 trials as the held-out unit; subject-grain LOO treats each of the 47 subjects as the held-out unit, the audit-relevant question (predictions on a new person). Both grains are reported in Table 3.

Three things to read off the table. First, the top three models (M2b, M3, M4) are tied at both grains: trial-grain $\Delta$elpd of 0.14 and 0.21 are smaller than their dse of 0.31 and 0.19, and subject-grain $\Delta$elpd of 0.96 and 2.30 are smaller than their dse of 2.64 and 2.85. Second, the rank order flips between grains (trial: M3 > M4 > M2b; subject: M2b > M3 > M4), but the audit conclusion does not depend on which grain wins: **the sex layer and the anthropometric layer add nothing detectable on top of Model 2b's structural baseline.** Third, the gap from the top three to the rest is large. Model 2 (Student-t but cohort $\sigma_{\text{obs}}$) is about 250 elpd worse on both grains; Model 2a (per-subject $\sigma_{\text{obs}}$ but Normal likelihood) is 645–736 elpd worse. Both the Student-t likelihood and the per-subject observation scale are doing work; neither is sufficient on its own.

| | Trial-grain | | | Subject-grain | | |
|---|---|---|---|---|---|---|
| Model | $\widehat{\text{elpd}}$ | $\Delta$elpd | dse | $\widehat{\text{elpd}}$ | $\Delta$elpd | dse |
| Model 3 (+ sex) | **−9,306.4** | 0 (ref) | 0 | −9,329.2 | +0.96 | 2.64 |
| Model 4 (+ anthropometrics) | −9,306.6 | +0.14 | 0.31 | −9,330.6 | +2.30 | 2.85 |
| Model 2b (subj. + per-subj. $\sigma$) | −9,306.7 | +0.21 | 0.19 | **−9,328.2** | 0 (ref) | 0 |
| Model 2 (subj. + cohort $\sigma$) | −9,561.8 | +255.4 | 30.3 | −9,578.0 | +249.7 | 58.2 |
| Model 1-t (Student-t, pooled) | −9,797.8 | +491.4 | 35.8 | −9,810.2 | +482.0 | 82.8 |
| Model 2a (per-subj. $\sigma$, Normal) | −10,043.1 | +736.6 | 72.1 | −9,973.9 | +645.6 | 137.4 |
| Model 1 (Normal, fully pooled) | −10,669.8 | +1,363.3 | 53.3 | −10,689.1 | +1,360.9 | 143.3 |

Table 3: PSIS-LOO comparison at trial and subject grain. Bold marks the best per grain; $\Delta$elpd is computed against each grain's own best model.

**Pareto-$k$ caveat.** Subject-grain LOO triggers Pareto-$k$ warnings on five of seven models (max-$k$ 1.26–1.45, with 39–45 of 47 subjects flagged $k > 0.7$). This is the leave-one-group-out PSIS failure mode documented in Vehtari et al. [2017] and Bürkner et al. [2021]: importance sampling from a single full-posterior fit cannot approximate the leave-one-out distribution of a parameter that is itself defined by the held-out group. The audit conclusion is robust to the caveat because the top three models are tied within dse at both grains. If subject-grain LOO is unreliable, the trial-grain conclusion still holds, and the two agree on which models tie. Subject-stratified $k$-fold cross-validation is the principled estimator and is left as future work.

# 7 Results

The headline result is in Figure 3: all five demographic-effect parameters (the raw sex contrast from Model 3, the residual sex contrast from Model 4, and the three anthropometric coefficients) have 95% HDIs that cross zero. Posterior summaries for both reporting models are in Table 4.

| | Model 3 | | Model 4 | |
|---|---|---|---|---|
| Parameter | Mean | 95% HDI | Mean | 95% HDI |
| $\alpha_{\text{sex},F}$ | −3.47 | [−4.86, −2.13] | −3.68 | [−5.30, −2.24] |
| $\alpha_{\text{sex},M}$ | −3.15 | [−4.23, −2.07] | −3.03 | [−4.17, −1.93] |
| $\Delta_{F-M}$ (sex contrast) | −0.32 | [−2.03, +1.17] | −0.65 | [−2.64, +1.11] |
| $\beta_{\text{leg}}$ (per SD) | | | −0.51 | [−1.52, +0.47] |
| $\beta_{\text{iad}}$ (per SD) | | | +0.03 | [−1.04, +1.05] |
| $\beta_{\text{mass}}$ (per SD) | | | −0.15 | [−1.25, +1.00] |
| $\sigma_{\text{sex}}$ | 2.00 | [0.00, 6.25] | 2.26 | [0.00, 6.68] |
| $\sigma_{\text{subj}}$ | 3.01 | [2.31, 3.83] | 3.06 | [2.31, 3.92] |
| $\nu$ (Student-t df) | 1.69 | [1.53, 1.86] | 1.69 | [1.53, 1.86] |
| $\beta_{\text{slow}}$ | 1.04 | [0.68, 1.36] | 1.04 | [0.70, 1.37] |
| $\beta_{\text{fast}}$ | −1.22 | [−1.55, −0.85] | −1.22 | [−1.55, −0.87] |

Table 4: Posterior summaries (mean and 95% HDI) for Models 3 and 4. The cohort intercept $\mu_{\text{pop}}$ is omitted because it becomes poorly identified once the sex layer is added (HDI widths 8–10 spm in M3 and M4); for a single cohort-level number, Model 2b's $\mu_{\text{pop}}$ posterior of −3.25 spm, 95% HDI $[-4.19, -2.36]$, is the cleanly identified estimate.

**Mean-level interpretation.** The mean cadence error is about −3 spm in both sex groups: the algorithm undercounts by roughly 3 spm on average for both female and male subjects. The sex-specific intercepts $\alpha_{\text{sex},F}$ and $\alpha_{\text{sex},M}$ are well identified, with HDI widths of about 2 spm. Their difference $\Delta_{F-M}$ is small

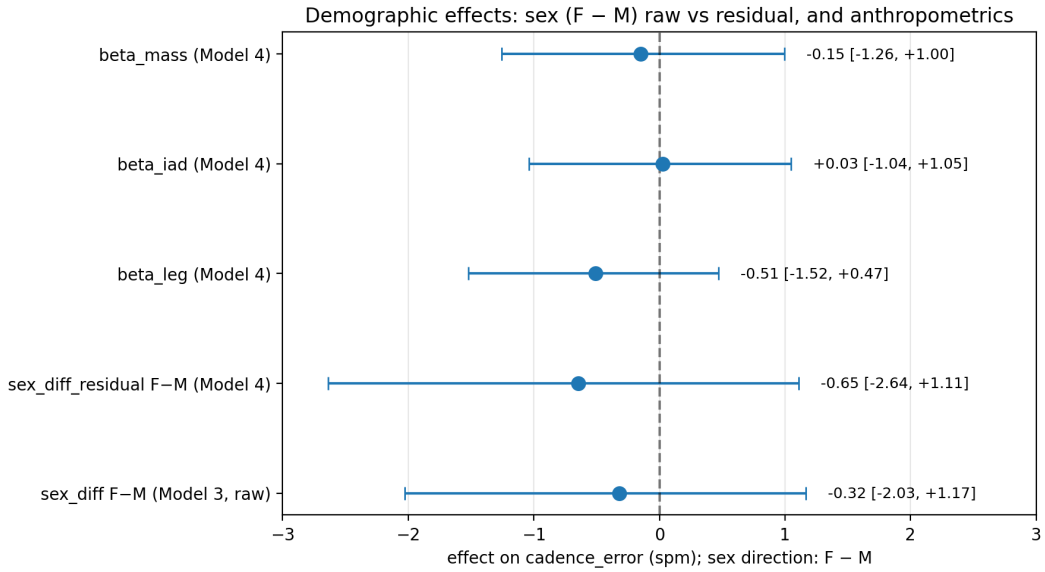


Figure 3: Posterior means and 95% HDIs for the five demographic-effect parameters. The raw F−M sex contrast (Model 3, bottom row) is $-0.32$ spm with HDI $[-2.03, +1.17]$; adjusting for measured leg length, inter-ASIS distance, and mass at the subject level (Model 4) widens the residual contrast to $-0.65$ spm with HDI $[-2.64, +1.11]$. Anthropometric coefficients are all credibly small with HDIs crossing zero. The $\beta_{\text{leg}}$ posterior is directionally consistent with the literature [Rowe et al., 2011] but not detectably distinct from zero. The opposite-direction shift between the two sex contrasts indicates suppression rather than mediation; see discussion below.

and credibly contains zero in both Models 3 and 4. Anthropometric coefficients in Model 4 are all credibly within $\pm 1.5$ spm per SD.

Compare this to the prior expectation from Rowe et al. [2011], which puts walking cadence variation at more than 20 spm across the adult height range: the algorithm's error is not modulated by body geometry on a scale anywhere near how cadence itself is modulated by body geometry.

**Suppression, not mediation.** The audit was set up to test for mediation: if anthropometrics explained the sex effect, $\Delta^{\text{resid}}_{F-M}$ in Model 4 should shrink toward zero relative to Model 3's $\Delta_{F-M}$. The posterior shows the opposite. Model 4's residual contrast is roughly twice the magnitude of Model 3's raw contrast ($-0.65$ vs. $-0.32$ spm, HDIs widening from $\pm 1.6$ to $\pm 1.9$ spm). The mechanism: in this cohort, sex and body geometry are correlated (sex-M with leg length $+0.31$, with mass $+0.36$), and the posterior signs of $\beta_{\text{leg}}$ and $\beta_{\text{mass}}$ are negative, so larger anthropometrics pull male subjects' modeled means downward and narrow the raw F−M gap. Conditioning on anthropometrics removes that pull, and the residual gap widens. Neither contrast reaches HDI-excludes-zero, so the claim is the sign-of-shift, not the direction of any one effect.

**The variance is the story, not the mean.** The dominant pattern in the cohort is between-subject heterogeneity, not between-sex heterogeneity. Per-subject posterior intercepts $\alpha_{\text{subj}}[s]$ span roughly 14 spm under the reporting models (Figure 4; full per-subject posterior in Appendix Table 5). Subject 31 (M) has the most negative posterior mean at $-12.6$ spm, with a 95% HDI that cleanly excludes both $\mu_{\text{pop}}$ and zero. The next four most-biased subjects (47, 16, 33, 17) span $-7.1$ to $-11.1$ spm. Both ends of the caterpillar are male: subject 31 at the low end, subjects 29 and 06 at the high end (both around $+1.5$ spm). The between-subject SD $\sigma_{\text{subj}} \approx 3$ spm is essentially unchanged across Models 2b, 3, and 4: adding the sex layer and three anthropometric predictors absorbed effectively zero between-subject variance. This is independent confirmation of the demographic null. If sex or body geometry had explanatory power on the per-subject

intercept, $\sigma_{\text{subj}}$ would shrink as those layers were added. It does not.

**The one detected structural effect is speed.** The trial-level speed coefficients $\beta_{\text{slow}} \approx +1.04$ spm and $\beta_{\text{fast}} \approx -1.22$ spm are stable across Models 2b, 3, and 4 with HDIs cleanly excluding zero. At slow speeds the IMU over-counts by about 1 spm relative to comfortable; at fast speeds it under-counts by about 1.2 spm. This is the one detectable structural bias in the cohort, in contrast to the undetectable demographic effects.

# 8 Discussion and limitations

The headline finding is a tightly bounded null on demographic effects on cadence error: the F−M sex contrast is constrained within roughly ±1.5 spm and per-SD anthropometric effects within roughly ±1.5 spm, against a literature prior [Rowe et al., 2011] of ∼20 spm cadence variation across the adult height range. The null direction replicates Abel et al. [2011]'s frequentist null on the same question, with posterior credible intervals instead of a non-rejection. The audit cannot rule out demographic effects in the broader population: the cohort's anthropometric range is narrower than a representative adult sample, and the F:M ratio is 36:64. What it can rule out, in this cohort and within its range, is demographic effects on cadence *error* of magnitude comparable to demographic effects on cadence *itself*. The mechanism prediction from Rowe et al. [2011] (longer legs → lower cadence at matched intensity → more undercount under a fixed-threshold detector) is directionally consistent with the $\beta_{\text{leg}}$ posterior mean of −0.51 spm per SD, but the HDI crosses zero.

The structural finding is that between-subject variation (∼14 spm range across 47 subjects) dwarfs any demographic contrast in this dataset. Straczkiewicz et al. [2023a]'s "one-size-fits-most" framing names the design intent; this audit shows the residual heterogeneity that frame leaves on the table is not predicted by the demographic averages it sets aside.

**Limitations.** The cohort is barefoot laboratory walking on one Finnish dataset; results do not transfer to free-living wear or to commercial proprietary detectors. The naive fixed-threshold detector is a lower bound on what a generic off-the-shelf algorithm achieves; tuned detectors and modern classifiers [Straczkiewicz et al., 2023b] would produce different residual patterns. Measured anthropometrics in the dataset do not include foot length or stride length, both of which would sharpen Model 4.

Within the present analysis, the continuous-outcome models estimate a mixture process (entry into the algorithm's failure mode plus within-failure severity) indirectly through a single likelihood. A two-stage model (logistic for failure-mode entry, continuous for severity given entry) is the natural extension and would account for the residual left-shoulder gap visible in Figure 5. Subject-grain PSIS-LOO is unreliable for leave-one-group-out with group-level parameters [Vehtari et al., 2017, Bürkner et al., 2021]; the audit conclusion is robust to this because the top three models tie within dse at both grains, but subject-stratified $k$-fold cross-validation is the principled estimator. Subject 31's outlier status is documented but not mechanistically investigated; the source of the ∼−12.6 spm mean undercount is not determined here.

# 9 Conclusion

A four-model hierarchical Bayesian audit of a naive threshold-based step-counting algorithm against motion-capture ground truth on the Kuopio Gait Dataset finds no detectable demographic structure in mean cadence error. The F−M sex contrast is bounded within ±1.5 spm (95% HDI), and per-SD anthropometric effects on leg length, inter-ASIS distance, and mass are similarly bounded, against a literature mechanism prediction roughly an order of magnitude larger. The substantive finding is that between-subject heterogeneity dominates demographic structure in this cohort. Sharpening this estimate would require a larger and de-

mographically wider cohort; extending it would require a two-stage mixture-process model that separately estimates entry into the algorithm's failure mode from severity within it.

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## Note on scope

This project was the final assignment for ISyE 6420 (Spring 2026) and doubles as a portfolio piece for a wearables data-science role, which shaped the scope. The full specification is a four-model hierarchical chain in PyMC, posterior-predictive-driven likelihood selection, and two-grain LOO cross-validation. Code, the frozen processed dataframe, and the build pipeline needed to regenerate every figure and table in this report are at `https://github.com/meeshmg/wearable-calibration-bayes`.

# A Appendix

**Per-subject posterior summary.** Table 5 reports the posterior mean and 95% HDI for $\alpha_{\text{subj}}[s]$ under Model 3, sorted most negative first. Generated from `idata_model_3.nc` via `az.summary` with `hdi_prob=0.95`; source CSV at `report_resources/appendix_per_subject_summary.csv`.

Table 5: Per-subject $\alpha_{\text{subj}}$ posterior summary, Model 3, sorted ascending by posterior mean (95% HDI). The full 47-row table is in the committed CSV; the first 10 rows are reproduced here.

| Subject | Sex | $n_{\text{trials}}$ | Mean (spm) | 95% HDI |
|---|---|---|---|---|
| 31 | M | 60 | −12.63 | [−15.21, −10.24] |
| 47 | F | 60 | −11.14 | [−14.97, −7.16] |
| 16 | F | 59 | −7.76 | [−11.13, −4.53] |
| 33 | F | 58 | −7.07 | [−10.78, −3.72] |
| 17 | M | 59 | −6.63 | [−8.33, −5.10] |
| 08 | F | 60 | −6.56 | [−9.90, −3.47] |
| 21 | M | 60 | −5.89 | [−8.72, −3.07] |
| 22 | M | 58 | −5.86 | [−7.52, −4.24] |
| 03 | M | 59 | −5.79 | [−8.66, −3.13] |
| 50 | M | 61 | −4.75 | [−5.31, −4.13] |
| *(rows 11–47 in committed CSV; remaining subjects' posteriors fall between −4.5 and +1.5 spm, with most subjects' HDIs containing $\mu_{pop} \approx -3.2$ spm and excluding zero on the negative side)* | | | | |

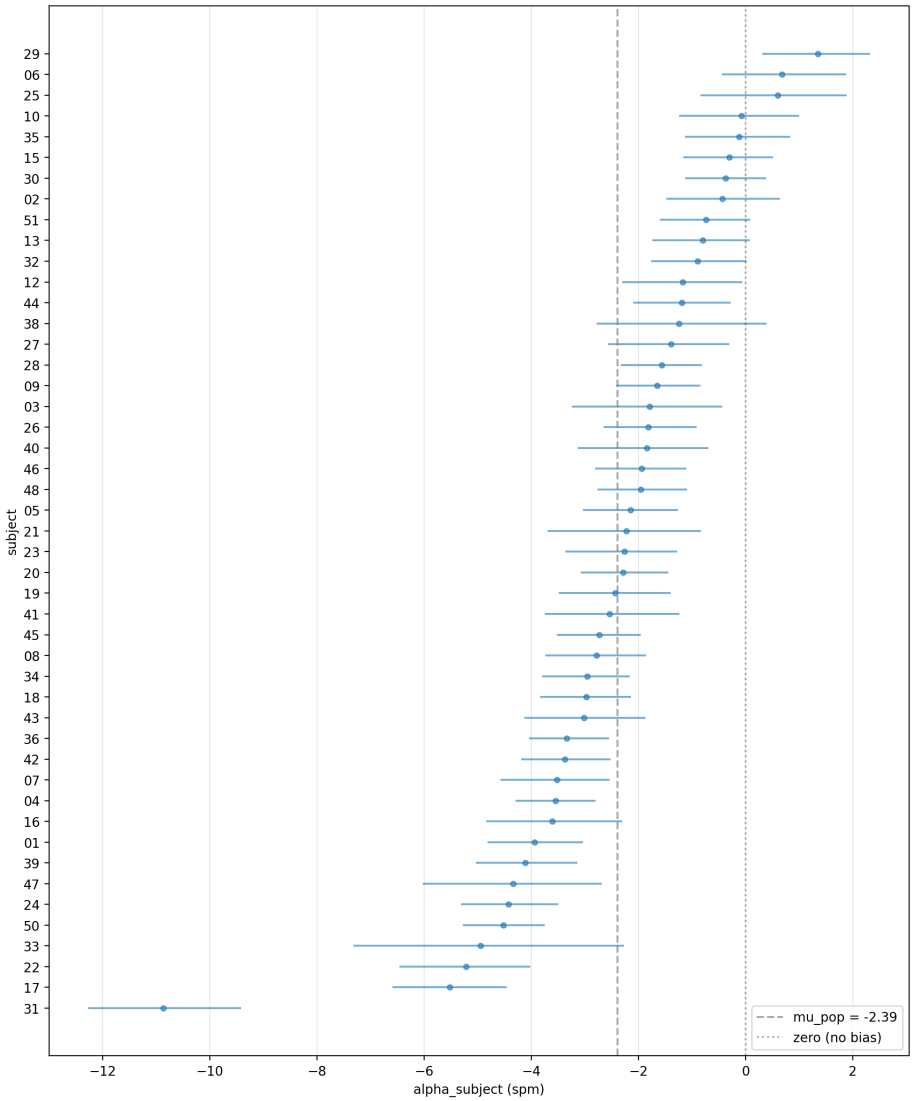


Figure 4: Per-subject posterior means and 95% HDIs for $\alpha_{\text{subj}}[s]$ from Model 2 (subject partial pooling with cohort observation scale), sorted ascending by posterior mean. Vertical dashed line: posterior mean of the population intercept $\mu_{\text{pop}} = -2.39$ spm. The figure is shown from Model 2 because that fit's caterpillar is the visual motivation for the per-subject observation scale adopted in Model 2b. Under the reporting models (M3 and M4), per-subject sigma absorbs trial-level erraticism and several subjects' intercepts shift further from $\mu_{\text{pop}}$: subject 31's posterior mean moves from $-10.9$ to $-12.6$ spm, subject 47's from $-4.4$ to $-11.1$ spm. The full per-subject posterior under the reporting model is in Appendix Table 5.

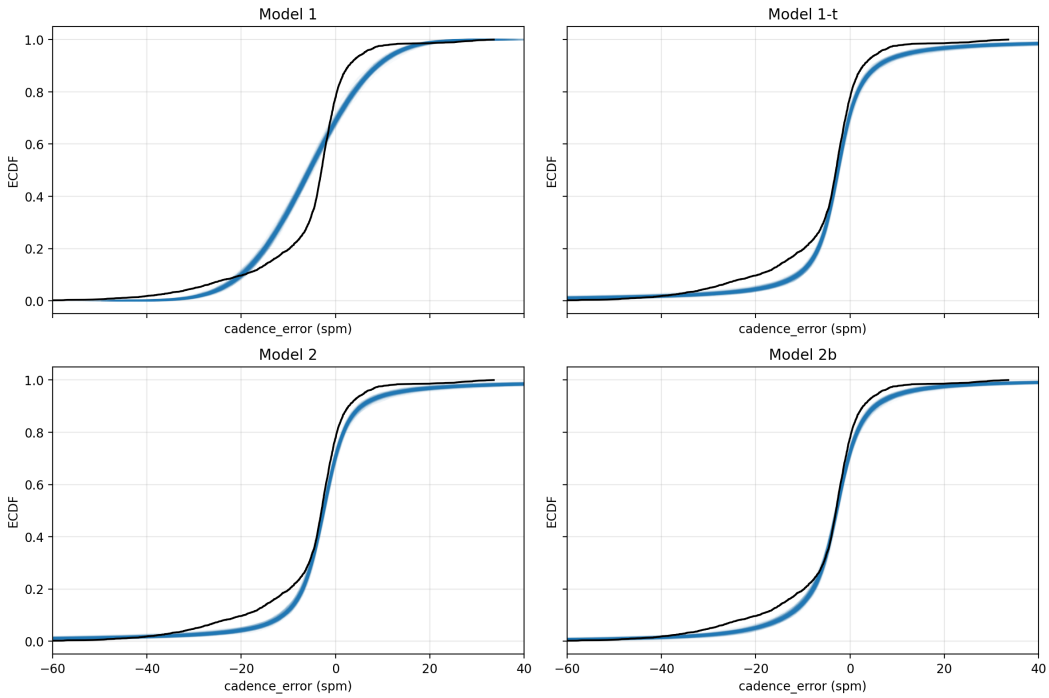


Figure 5: Posterior predictive cumulative distribution functions across the four likelihood-and-scale specifications. Black: observed ECDF of cadence error ($n = 2{,}781$ trials). Blue band: 200 posterior predictive draws per panel. Top-left (Model 1, Normal, fully pooled): observed ECDF exits the band on both tails, prompting the Student-t refit. Top-right (Model 1-t, Student-t, fully pooled): bulk fit improves but the deep left tail still sits above the predictive band. Bottom-left (Model 2, Student-t with cohort observation scale and subject partial pooling): subject intercepts do not shift the tail shape; posterior $\nu$ in fact decreases relative to M1-t (1.11 vs. 1.19), indicating the fat tails are trial-level. Bottom-right (Model 2b, Student-t with per-subject observation scale $\sigma_{\text{obs}}[s]$): tail fit tightens substantially; right-tail overshoot resolves; residual left-shoulder under-coverage in the $-20$ to $-10$ spm range remains and is the unmodeled mixture process named in the limitations.